

00:52

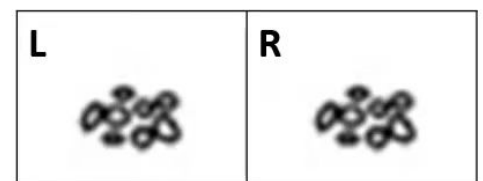
## Vacuum World Problem

There are two dirty square blocks and we need to clean these dirty square blocks by using vacuum cleaner. *Agent*

• Percepts : Location and content e.g. (L,Dirty)

• Possible actions for Vacuum cleaner

1. Move Left (L)
2. Move Right (R)
3. Suck (S)
4. No Operation (NoOp)



01:16

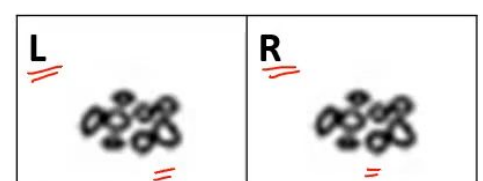
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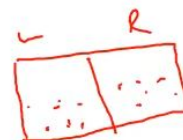
02:23

## Vacuum World Problem...

- Initial State
- Vacuum Agent can be in any states (Left or Right) shown in the picture
- Successor function
- Successor function generates legal states resulting from applying the three actions {Left, Right, and Suck}
- The states space is shown in the picture, there are 8 world states
- Goal test
- Checks whether all squares are clean
- Path cost
- Each step costs 1, so the path cost is the sum of steps in the path from Initial state to goal state.

02:58

## Vacuum World Problem...



- Initial State ✓
- Vacuum Agent can be in any states (Left or Right) shown in the picture
- ✓ Successor function
- Successor function generates legal states resulting from applying the three actions {Left, Right, and Suck}
- The states space is shown in the picture, there are 8 world states
- ✓ Goal test
- Checks whether all squares are clean

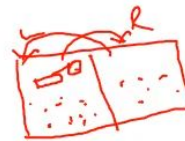
Checks whether all squares are clean.

#### ✓ Path cost

- Each step costs 1, so the path cost is the sum of steps in the path from Initial state to goal state.

03:43

## Vacuum World Problem...



#### ✓ Initial State ✓

- Vacuum Agent can be in any states (Left or Right) shown in the picture

#### ✓ Successor function

- Successor function generates legal states resulting from applying the three actions {Left, Right, and Suck}
- The states space is shown in the picture, there are 8 world states

#### ✓ Goal test

- Checks whether all squares are clean

#### ✓ Path cost

- Each step costs 1, so the path cost is the sum of steps in the path from Initial state to goal state.

03:47

## Vacuum World Problem...



#### ✓ Initial State ✓

- Vacuum Agent can be in any states (Left or Right) shown in the picture

#### ✓ Successor function

- Successor function generates legal states resulting from applying the three actions {Left, Right, and Suck}
- The states space is shown in the picture, there are 8 world states

✓ Goal test

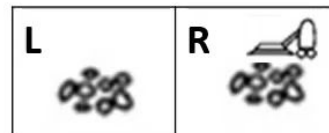
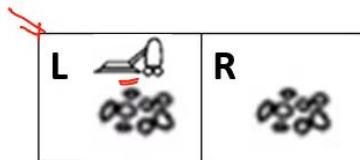
- Checks whether all squares are clean ✓

✓ Path cost

- Each step costs 1, so the path cost is the sum of steps in the path from Initial state to goal state.

05:10

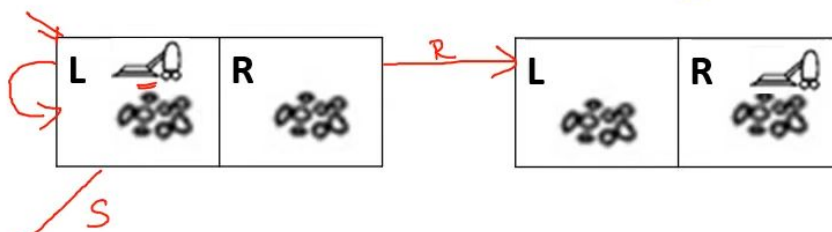
## Vacuum World – Transition diagram



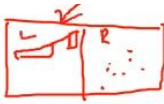
2 ← block  
3 ← action  
2<sup>3</sup> ← 8, Start

05:56

## Vacuum World – Transition diagram

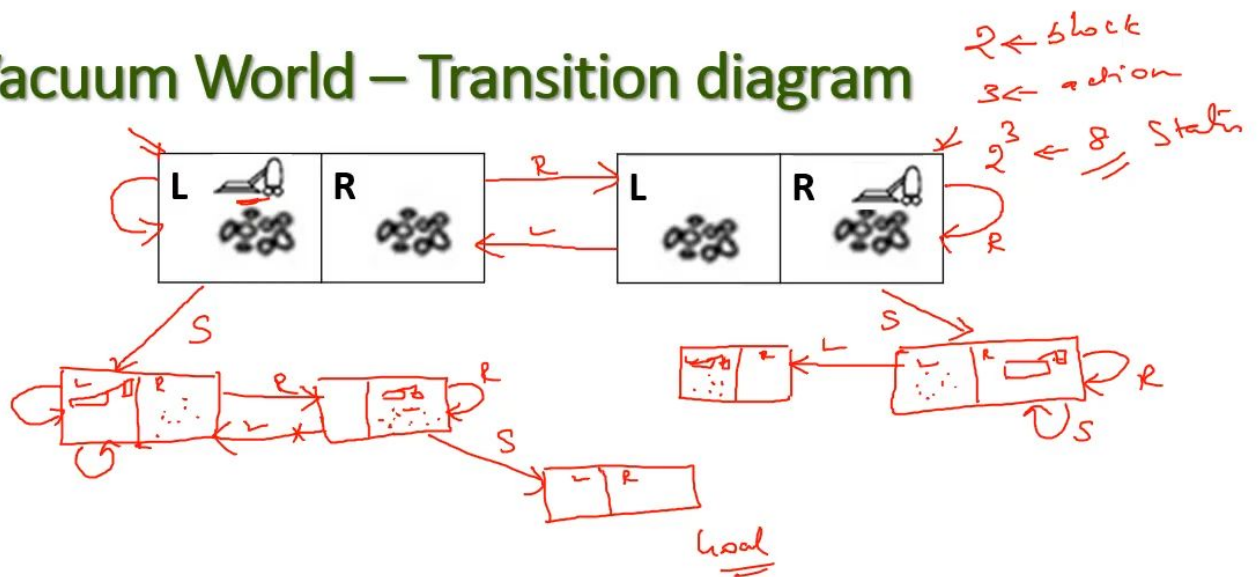


2 ← block  
3 ← action  
2<sup>3</sup> ← 8, Start



09:09

## Vacuum World – Transition diagram

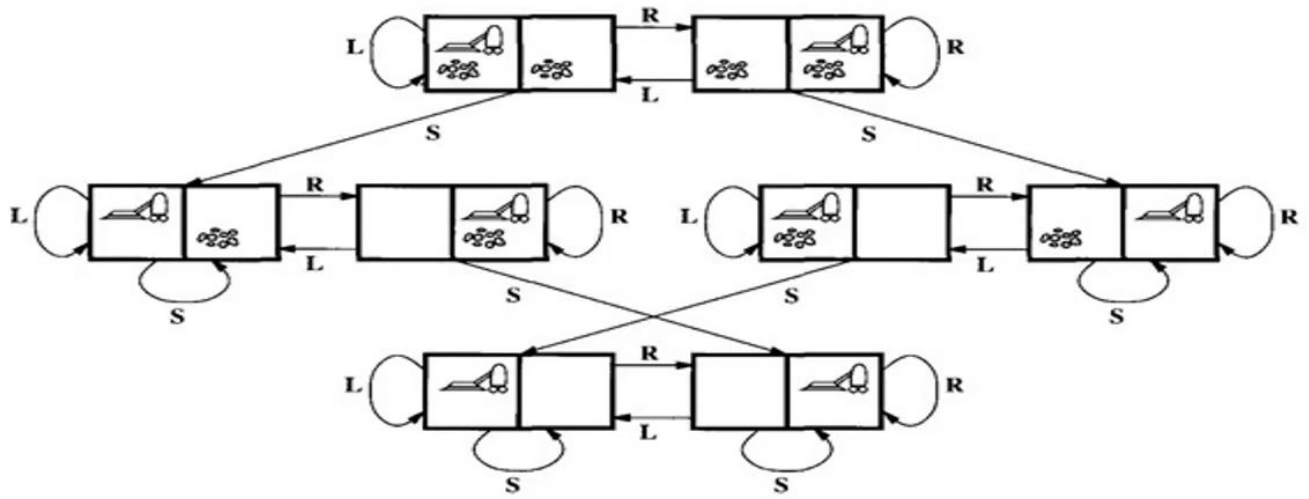


10:29

## Transition Diagram for Vacuum World Problem

- Path cost is 3 in our problem.

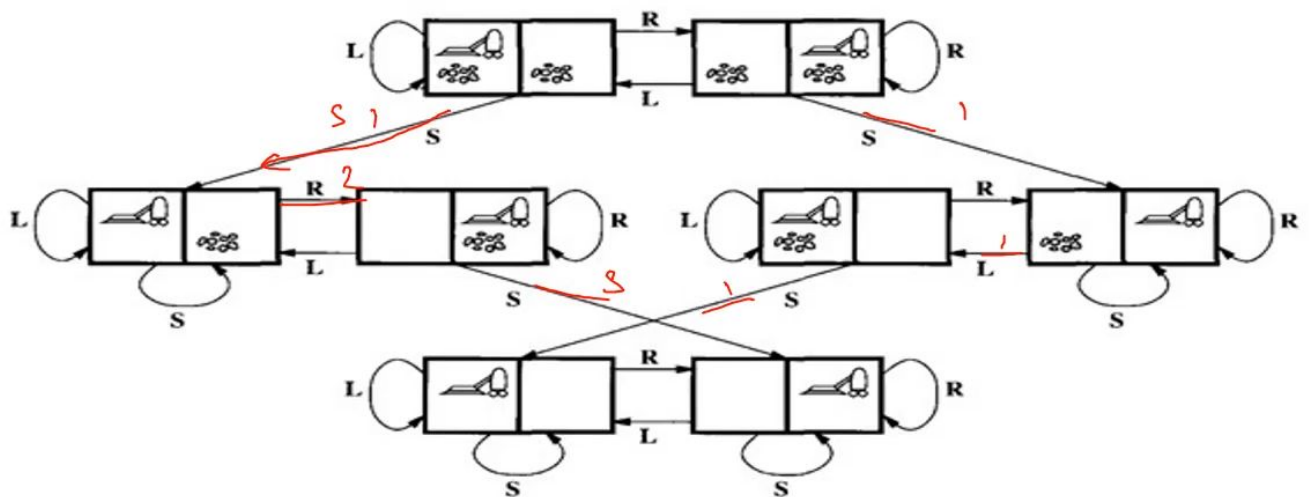




11:22

## Transition Diagram for Vacuum World Problem

- Path cost is 3 in our problem. *1 ← action*



11:24

Thank You

# Thank you

- **Toy problems - Vacuum World Problem**
- **Initial state, successor function, goal test and path cost**
- **Transition Diagram**
- Up next...
- 8 puzzle problem.















